CBSE Class 10 SCIENCE - Board Exam 2017 (Set-1, Code 31/1)

CHAPTER-WISE QUICK NOTES (30 minutes)

Read the chapter concepts, then open the 2017 paper and write the answers yourself.

• 0-22 min: read calmly and understand each concept. 22-27 min: revise the 6 high-yield facts: 10% law (Q3), ethanol dehydration (Q7), Na/Mg/Al logic (Q10), f = 1/P (Q16), Mendel Tt x Tt (Q14), myopia correction (Q24). Last 3 min: practise Q35 Amoeba stages + Q17 prism diagram.

• Green lines = exact keywords/answers the examiner looks for - use those words in your paper.

• Wherever a block says [Draw diagram] (Q16, 17, 22, 23, 24, 35): a neat, labelled diagram alone fetches marks.

CHAPTER 1: CARBON AND ITS COMPOUNDS

Covers exam questions: Q1, Q7, Q8, Q19, Q25, Q26, Q27, Q34 (1+3+3+5+1+1+1+2 = 17 marks)

1.1 Hydrocarbons & homologous series

Hydrocarbons = compounds made of ONLY carbon and hydrogen. A homologous series is a family of compounds having the same general formula and similar properties; successive members differ by -CH2-.

Series starting with methane = ALKANES, general formula CnH2n+2 : CH4 (methane), C2H6 (ethane), C3H8 (propane). Alkenes: CnH2n (ethene C2H4, C=C double bond). Alkynes: CnH2n-2 (ethyne C2H2, carbon-carbon triple bond C≡C).

Write in exam: Q1 - 2nd member C2H6, 3rd member C3H8. Q19 - definition + general formulas + draw structures of CH4, C2H4, C2H2.

1.2 Ethanol and its dehydration

Structural formula of ethanol: CH3-CH2-OH (two carbons, one of them carries the -OH group).

On heating ethanol with EXCESS conc. H2SO4 at 443 K, a water molecule is removed and ethene is formed:

CH3-CH2-OH --(Conc. H2SO4, 443 K)--> CH2=CH2 + H2O

Write in exam (Q7): Ethene is formed; conc. H2SO4 acts as a DEHYDRATING AGENT (removes water).

1.3 Esterification vs Saponification

Esterification = carboxylic acid + alcohol -> ester + water: CH3COOH + C2H5OH -> CH3COOC2H5 + H2O. Esters have sweet fruity smell, so used in perfumes and artificial flavours.

Saponification = ester + NaOH -> alcohol + soap (sodium salt of the acid): CH3COOC2H5 + NaOH -> C2H5OH + CH3COONa. Used to prepare soap.

Write in exam (Q8): the two definitions + both equations + one use each (perfume/flavour; making soap). Trick: ester BANANA = esterification; ester se SOAP = saponification.

1.4 Converting alkene to alkane (Hydrogenation)

Adding hydrogen to an unsaturated hydrocarbon (double/triple bond) converts it into a saturated one. Needs Nickel (Ni) catalyst + heat:

CH2=CH2 + H2 --(Ni, heat)--> CH3-CH3

Write in exam (Q19): name of reaction = hydrogenation (addition reaction); conditions = Ni catalyst + heat, with this equation.

1.5 Lab practicals (Section B)

Acetic acid + sodium bicarbonate: brisk bubbles of CO2 gas which is COLOURLESS and ODOURLESS. CH3COOH + NaHCO3 -> CH3COONa + H2O + CO2.

Q25 answer = (D)

Saponification is an EXOTHERMIC reaction - the outer surface of the beaker becomes HOT.

Q26 answer = (C)

Hard water contains dissolved salts of Ca2+ / Mg2+. So calcium sulphate and calcium chloride dissolved in distilled water make it hard.

Q27 answer = (D)

Preparing soap in lab: heat vegetable oil/fat with 20% NaOH solution while stirring; add NaCl (common salt) to separate the soap.

Nature test of the mixture: dip RED litmus paper in it -> turns BLUE (or phenolphthalein turns pink) => mixture is ALKALINE because excess NaOH remains.

Write in exam (Q34): oil/fat + NaOH solution as materials; red-litmus-turns-blue test = alkaline.

CHAPTER 2: PERIODIC CLASSIFICATION OF ELEMENTS

Covers: Q9, Q10 (3+3 = 6 marks)

2.1 Modern Periodic Table - facts & trends

The Modern Periodic Table has 7 PERIODS (horizontal rows) and 18 GROUPS (vertical columns).

Metallic character = tendency of an atom to LOSE electrons. Across a period (left to right) it DECREASES because nuclear charge increases and holds electrons more tightly. Down a group it INCREASES because atomic size increases, so outer electrons are lost easily.

Write in exam (Q9): 7 periods, 18 groups; trend (i) decreases across period, (ii) increases down a group; with these reasons.

2.2 Na, Mg and Al (Period 3, groups 1, 2, 13)

Electronic configurations: Na = 2,8,1 | Mg = 2,8,2 | Al = 2,8,3

Valency: Al loses all 3 outer electrons -> highest valency (3). Atomic radius: size decreases left to right -> Na is largest. Chemical reactivity of metals: decreases left to right -> Na is most reactive.

Write in exam (Q10): (a) Al - highest valency; (b) Na - largest atomic radius; (c) Na - maximum chemical reactivity; justify each with the trend/reason above.

CHAPTER 3: HOW DO ORGANISMS REPRODUCE

Covers: Q2, Q11, Q12, Q13, Q20, Q35 (1+3+3+3+5+2 = 17 marks)

3.1 What happens to DNA when a cell reproduces

Before a cell divides, its DNA first REPLICATES (makes two copies of itself). The two copies are then distributed equally between the two daughter cells.

Write in exam (Q2): DNA first replicates into two copies, which pass equally into the two daughter cells - so offspring get the same DNA as the parent.

3.2 Why is reproduction so important (3 reasons)

(1) It maintains continuity of the species (prevents extinction). (2) It maintains/increases the number of organisms. (3) It produces variations - and variations make EVOLUTION possible.

Write in exam (Q11): these three reasons, in one sentence each.

3.3 Vegetative propagation

Definition: a new plant grows from a vegetative part (root, stem or leaf) of the parent - no seeds involved. Methods: cutting, layering, grafting, tissue culture.

Advantages: (1) quick and easy; even SEEDLESS plants/fruits can be grown; (2) all good traits of the parent are preserved exactly. Disadvantages: (1) no variation is produced; (2) overcrowding, and diseases pass on from generation to generation.

Write in exam (Q12): definition + 2 advantages + 2 disadvantages.

3.4 Contraception (preventing pregnancy)

Three techniques developed: (1) Barrier method - condom / diaphragm; (2) Chemical method - oral contraceptive pills; (3) IUCD - Copper-T placed in the uterus. (Surgical: vasectomy in males, tubectomy in females.)

Copper-T is NOT meant for males (it sits inside the female uterus).

Impact on the family: proper spacing between children and a smaller family protect the mother's health; more resources per child make the family healthier and more prosperous.

Write in exam (Q13): name the 3 techniques + Copper-T (not for males) + the health/prosperity impact.

3.5 Human female reproductive system + placenta

Ovary: produces the ova (female gametes) and secretes the hormones estrogen and progesterone. Uterus: the embryo implants here; it protects and nourishes the growing foetus. Fallopian tube: carries the ovum towards the uterus and is the SITE OF FERTILISATION.

Placenta: a disc-like structure embedded in the uterus wall, connected to the foetus by the umbilical cord; it has VILLI giving a large surface area. Functions: supplies O2 and nutrients from the mother's blood to the foetus and carries away CO2 and excretory wastes from the foetus.

Write in exam (Q20): functions of ovary / uterus / fallopian tube + placenta structure & function as above.

3.6 Binary fission in Amoeba - 4 stages [Draw diagram]

Draw four stages in sequence, joined by arrows: (1) the nucleus elongates; (2) the nucleus divides into two; (3) two nuclei move apart and a constriction appears in the middle of the cell; (4) the cytoplasm divides -> two daughter Amoebae.

Write in exam (Q35): neat labelled 4-stage diagram in the above order.

CHAPTER 4: HEREDITY AND EVOLUTION

Covers: Q14, Q15, Q21, Q28, Q29 (3+3+5+1+1 = 13 marks)

4.1 Mendel: how a trait can be inherited but not expressed

Mendel crossed Tt x Tt (F1 self-pollination). In F1 all plants were TALL - the recessive dwarf trait (t) stayed hidden but was still present inside (Tt). In F2, 25% plants were dwarf (tt) again.

Conclusion: a recessive trait IS inherited (present in the genes of F1) but NOT expressed (not visible). It can pass silently to later generations and reappear.

Write in exam (Q14): the Tt x Tt cross + this conclusion.

4.2 Evolution and classification are interlinked

Organisms showing more similarities share a closer common ancestor. Since classification groups organisms by their similarities and differences, it actually displays their evolutionary relationships. So the two are interlinked.

Write in exam (Q15): this statement + reason + a small example (e.g., all mammals share basic body plan -> common ancestor).

4.3 Acquired traits vs Inherited traits

Inherited traits: present since birth, carried in the DNA/genes received from parents - eye colour, blood group, free or attached earlobe. Acquired traits: developed during one's lifetime - skill of swimming, a muscular body from exercise, knowledge of a language.

Acquired traits are NOT inherited because they do not change the DNA of the germ cells (gametes). Only changes in gamete-DNA can pass to the next generation.

Example: the muscular body of a wrestler/bodybuilder is not born in his child automatically.

Write in exam (Q21): distinguish with one example each + the gamete-DNA reason + example.

4.4 Homologous structures & seed types (Section B MCQs)

Homologous structures = organs with the same basic origin/structure (may perform different functions). Radish and carrot are BOTH modified ROOTS -> homologous. (Potato is a stem, not a root; tomato is a fruit.)

Q29 answer = (C) Radish and Carrot

Dicot seeds have TWO cotyledons: gram (chana), pea (matar), ground-nut (moongphali). Wheat, maize and barley are MONOCOTS (one cotyledon).

Q28 answer = (B) Gram, Pea and Ground-nut

CHAPTER 5: LIGHT - REFLECTION AND REFRACTION

Covers: Q4, Q16, Q22, Q23, Q30, Q31, Q36 (2+3+5+5+1+1+2 = 19 marks)

5.1 Concave lens - image is always virtual, erect, diminished

A CONCAVE lens always forms a virtual, erect and diminished image, between the focus (F1) and the optical centre, whatever the object position.

Power-focal length link: P = 1/f (f in metres). For concave lens P is NEGATIVE. If numerical value of P = 10 D: P = -10 D, f = 1/P = -0.1 m = -10 cm.

Write in exam (Q16): nature = concave lens + ray diagram + f = -10 cm (Cartesian sign convention).

Typical numerical: f = -15 cm, u = -30 cm -> lens formula 1/v - 1/u = 1/f gives v = -10 cm.

Write in exam (Q4): image is (1) virtual (2) erect (3) diminished (4) formed between focus and optical centre (10 cm, same side as object).

5.2 Convex mirror - image always diminished, erect, virtual

A CONVEX mirror always forms a diminished, erect, virtual image behind the mirror, between pole (P) and focus (F). Use: rear-view mirror in vehicles because (i) it gives a wide field of view and (ii) the image is erect.

Radius of curvature (R) = radius of the sphere of which the mirror forms a part. f = R/2. R = +24 cm (positive sign => convex mirror); f = +12 cm.

Write in exam (Q23): type = convex mirror + ray diagram + use & reason; R definition; f = +12 cm, convex.

5.3 Convex lens observation table (u-v) [Draw diagram]

Key: when |u| = |v| = 2f. Observation 3 shows u = -40 cm, v = +40 cm => object at 2F => 2f = 40 => f = +20 cm.

Wrong observation: S.No. 6, because u = -15 cm lies INSIDE f (20 cm); then the image must be virtual with v NEGATIVE - but the table shows v = +120 cm, which is impossible.

For obs 2: u = -60 cm (beyond 2F) -> image forms between F and 2F (v = +30 cm); magnification m = v/u = 30/60 = 0.5 -> image is inverted and half the object size. Draw the ray diagram with scale 1 cm = 10 cm.

Write in exam (Q22): f = +20 cm (justify via obs 3 / lens formula); wrong = S.No. 6 (reason above); ray diagram of obs 2 with m = 0.5.

5.4 Practical lens/mirror questions (Section B)

A device that converges a parallel beam onto a screen behind it (20 cm) is a CONVEX lens with f = 20 cm; a MIRROR that converges a parallel beam onto a screen placed in FRONT of it (25 cm) is a CONCAVE mirror with f = 25 cm.

Q30 answer = (D)

If the image of a distant object is blurred, the lens should be moved towards the screen OR away from it - it depends on the object's position.

Q31 answer = (D)

Candle at 2 m from a convex lens (f = 10 cm), then candle moved slowly towards the lens: (A) move the lens AWAY from the screen (image distance v increases as u decreases); (B) size of the image INCREASES; (C) brightness/intensity DECREASES (same light spread over a larger image); (D) at 5 cm (less than f = 10 cm) NO image is obtained on the screen - a virtual image cannot be caught on a screen.

Write in exam (Q36): these four points, one line each.

CHAPTER 6: THE HUMAN EYE AND THE COLOURFUL WORLD

Covers: Q17, Q24, Q32, Q33 (3+5+1+1 = 10 marks)

6.1 Dispersion by a prism + Newton's two-prism experiment

Cause of dispersion: white light consists of 7 colours; each colour refracts (bends) by a DIFFERENT amount while passing through glass - violet bends the most, red the least - so white light splits into a spectrum.

Newton took a second identical prism placed INVERTED next to the first: the seven colours recombined into white light. This proved white light is made of seven colours.

Write in exam (Q17): the cause + Newton's recombination experiment + [Draw diagram] prism 1 -> spectrum; inverted prism 2 -> white beam.

6.2 Myopia (near-sightedness) and its correction [Draw diagrams]

In myopia, a person sees nearby objects clearly but distant objects (> 5 m here) look blurred because their image forms IN FRONT of the retina.

Two possible causes: (1) the eyeball has become too long; (2) the eye lens has too much curvature (focal length too short).

Correction: use a CONCAVE (diverging) lens; it spreads the rays slightly so the image shifts back exactly onto the retina.

Power: f = -5 m => P = 1/f = -0.2 D (negative, as per new Cartesian sign convention, for a concave lens).

Write in exam (Q24): 2 causes; two ray diagrams - (i) image before retina, (ii) concave lens -> image on retina; P = -0.2 D.

6.3 Glass slab & prism angles (Section B MCQs)

Glass slab: the emergent ray comes out parallel to the incident ray, so angle of incidence i = angle of emergence e. Inside the glass the ray bends TOWARDS the normal, so refraction angle r is smaller. Hence i = e > r.

Q32 answer = (B)

Prism diagram labels: p = angle of incidence (between the incident ray and the normal), y = angle of emergence (between emergent ray and the normal), z = angle of deviation (between the extended incident ray and the emergent ray). So incidence, emergence, deviation = p, y, z.

Q33 answer = (C)

CHAPTER 7: OUR ENVIRONMENT + MANAGEMENT OF NATURAL RESOURCES

Covers: Q3, Q5, Q6, Q18 (1+2+2+3 = 8 marks)

7.1 10% energy law (food chain)

At each trophic level of a food chain, only about 10% of the energy passes to the next level. If the lion gets 100 J, multiply by 10 for each step backwards: deer had 1000 J, plants (producers) had 10,000 J.

Write in exam (Q3): producer (plants) = 10,000 J.

7.2 Conserving forests and wildlife

Forests: (i) they maintain rainfall and groundwater and prevent soil erosion & floods; (ii) provide O2, wood, medicines and many forest products. Wildlife: (i) maintains ecological balance and food chains; (ii) source of medicines, gene pool and tourism.

Write in exam (Q5): any two advantages for forests + any two for wildlife.

7.3 Water harvesting & saving water

Community-level water harvesting: (1) recharges groundwater so wells/ponds do not dry up; (2) reduces flooding and provides water even in the dry season.

To stop the underground water table from falling further: RAINWATER HARVESTING - rooftop rainwater is led through pipes into a recharge pit/well so that it soaks into the ground and raises the water table.

Awareness ideas for the neighbourhood: posters/slogans, nukkad natak (street play), door-to-door campaign, a talk in the school/colony.

Write in exam (Q6, Q18): two community advantages; any two awareness ways; rainwater harvesting with explanation.

Total: Section A = 75 marks + Section B = 15 marks = 90 marks

All 36 questions of the 2017 Set-1 paper are covered above. Now open the paper and start writing - Best of luck!